Machine Learning

01-05-26

PCA: Principle component Analysis

- unsupervised ML
- used for dimensionality reduction of data

PCA tries to solve the curse of dimensionality

- high dimensional data (data having too many features) has curse of dimensionality

- this type of data is not suitable for model training

too many features → high chances of overfiting
→ very slow training

original data

$x_1 \; x_2 \; x_3 \; \ldots \; x_{20}$ → $\boxed{\text{PCA}}$ → $P_1 \; P_2 \; P_3 \; \ldots \; P_{20}$

original features

principle components

↓

select top K most important PCs

if $K = 5$

$P_1 \; P_2 \; P_3 \; P_4 \; P_5$

↓

train model on these as features

$x_1 \; x_2 \; x_3 \cdots x_{20} \rightarrow \boxed{\text{ML model}} \rightarrow \hat{y}$

After PCA with $k = 5$

$p_1 \; p_2 \; p_3 \; p_4 \; p_5 \rightarrow \boxed{\text{ML model}} \rightarrow \hat{y}$

- Even if we are reducing the dimensions of the data, we don't want to loose much information
- we want to preserve as much information as possible.

How does PCA preserve information?

information in data is proportional to its variance.

eg.

d_1:	14	14	14	13	14	14	13	14	14	13	14
d_2:	14	15	100	2	4	6	-7	-8	5	0	1

low variance $\rightarrow$ less information
high variance $\rightarrow$ more information

Objective of PCA is to reduce the dimensions of data while preserving maximum variance

preserving variance = preserving information

maximum variance of the data occurs along its eigen vectors.

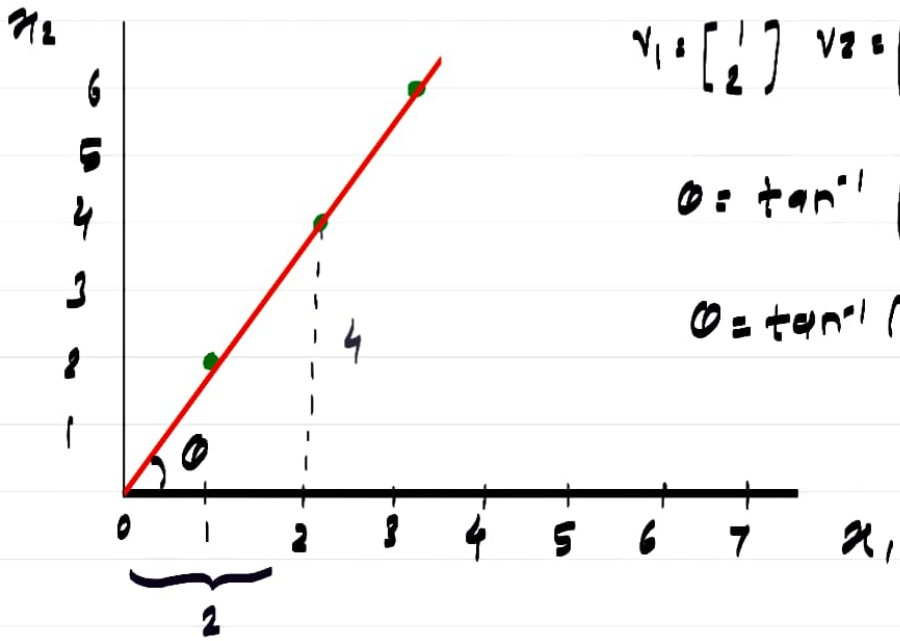
Lets understand eigen vectors :

Unit vector :

- vector having magnitude : 1

$$|u| = 1$$

- unit vectors are used to represent directions in N-dimensional space.



$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $v_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ $v_3 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$

$$\theta = \tan^{-1}\left(\frac{x_2}{x_1}\right)$$

$$\theta = \tan^{-1}(2)$$

$|v_1| = \sqrt{5}$ $|v_2| = \sqrt{20}$ $|v_3| = \sqrt{45}$

v_1 v_2 v_3 have different magnitudes but same direction

A unit vector can be used to represent their direction

$$u = \frac{v_1}{|v_1|} = \frac{v_2}{|v_2|} = \frac{v_3}{|v_3|}$$

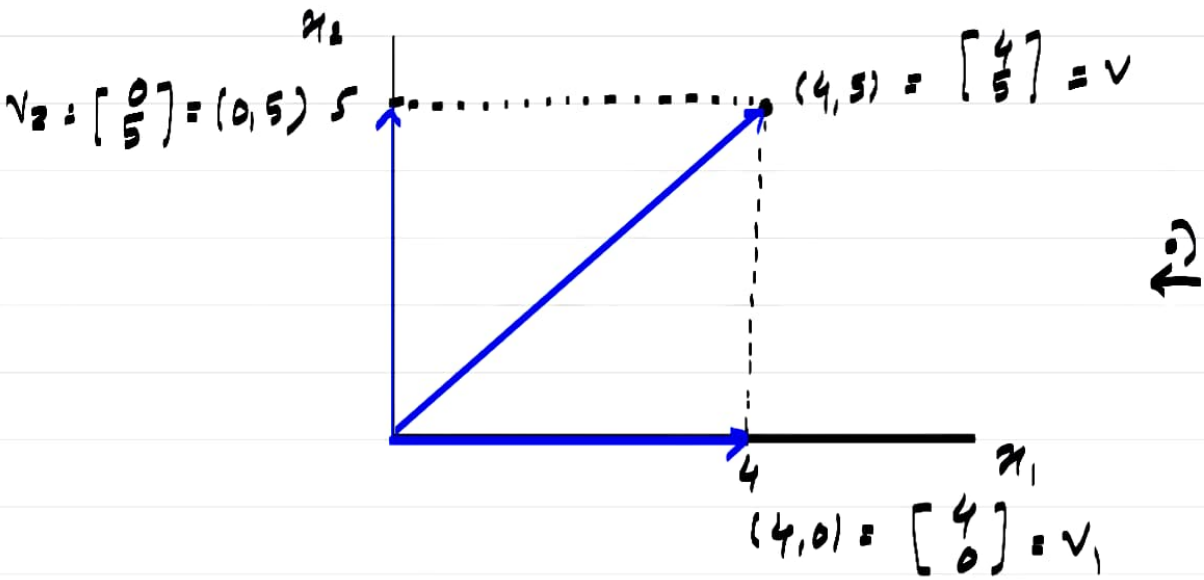
$$u = \frac{v_1}{|v_1|} = \frac{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}{|v_1|} = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$$u = \frac{v_2}{|v_2|} = \frac{\begin{bmatrix} 2 \\ 4 \end{bmatrix}}{\sqrt{20}} = \begin{bmatrix} 2/\sqrt{20} \\ 4/\sqrt{20} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

vectors having same direction also have the same unit vector

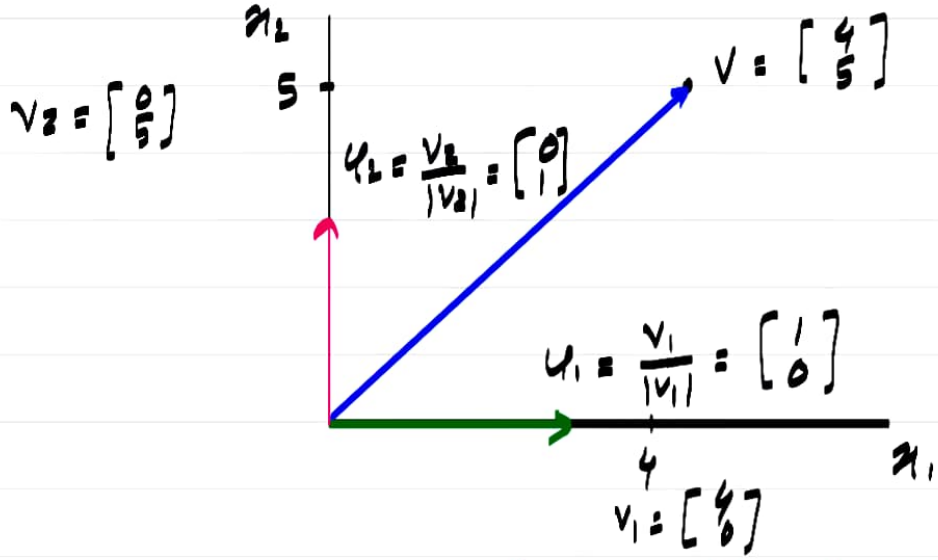
$$u = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} \quad \text{then} \quad |u| = \sqrt{\frac{1}{5} + \frac{4}{5}} = \sqrt{1} = 1$$

Projection of vector :



v_1 is projection of vector v along x_1 axis

v_2 is projection of vector v along x_2 axis



To find projection of a vector to any given direction we can use unit vector representing that direction

$$\text{proj } A \text{ on } B = (A.B)\frac{B}{|B|^2}$$

if B is a unit vector then $|B|^2 = 1$

$$\text{proj } A \text{ on } B = (A.B)\, B$$

for any unit vector u

$$\text{proj } A \text{ on } u = (A.u)\, u$$

for above example : $v = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ $u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\text{proj } v \text{ on } u_1 = (v.u_1)\, u_1$$

$$= (4+0)\begin{bmatrix} 1 \\ 0 \end{bmatrix} = 4 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix} = v_1$$

$$\text{proj } v \text{ on } u_2 = (v.u_2)\, u_2$$

$$= (0+5)\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \end{bmatrix} = v_2$$

Objective of PCA:

preserve maximum variance $\Longrightarrow$ eigen vectors give the direction for maximum variance

- First we find eigen vectors then we project the data along with their direction
- This projection transforms original data into principle components.

Eigen vectors and Eigen values:

matrix vector multiplication

$$w = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \qquad v_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$v_1' = w v_1 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2\times1 + 1\times3 \\ 1\times1 + 2\times3 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 0 \\ 4 \end{bmatrix} \text{ then } \quad v_2' = w v_2 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \end{bmatrix}$$

$$v_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad v_3' = w v_3 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3 v_3$$

$$v_3' = w v_3 = 3 v_3$$

$v_4 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ $\quad v_4' = Wv_4 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1\, v_4$

$v_4' = 1 . v_4$

$v_5 = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$ $\quad v_5' = Wv_5 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$

$v_5' = 1 . v_5$

for a unit vector v and matrix A
iff

$$Av = \lambda v$$

then v is called an eigen vector of matrix A
and
λ is the eigen value associated to given eigen vector

in previous example

$Wv_3 = \lambda_1 v_3 \qquad Wv_4 = \lambda_2 v_4 \qquad Wv_5 = \lambda_2 v_5$

$v_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ unit vector in direction of v_3 $= \dfrac{v_3}{|v_3|} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \leftarrow u_1$

$v_4 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ unit vector in direction of v_4 $= \dfrac{v_4}{|v_4|} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} \leftarrow u_2$

for vector v in direction of u_1
$Wv = 3v$
for vector v in direction of u_2
$Wv = 1 \times v$

u_1 and u_2 are eigen vector with eigen values 3 and 1 respectively

if V is an eigen vector of matrix A then

$$AV = \lambda V$$

where λ is a scaler

and

for any other vector V' in direction of V

$$AV' = \lambda V'$$

$$\underbrace{A}_{\text{matrix}}\underbrace{V}_{\text{vector}} = \underbrace{\lambda}_{\text{scaler}}\underbrace{V}_{\text{vector}}$$